LAB 09

# Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

| Item | Detail |
|---|---|
| Topic | Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca |
| Pipeline stage | Evaluation |
| TSC reference | A3, K3 |
| Duration | 25 min |
| Folder | labs/Lab09-performance-metrics/ |
| Files | Lab09-performance-metrics/evaluate_stage.py, pipeline/evaluate.py, common/metrics.py, Lab09-performance-metrics/solution/evaluate.py |
| Simulator page | http://localhost:8000/#/analytics |
| Mode | Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only. |

*Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.*

## Scenario

The investment committee wants to know which of the pipeline's signals deserves capital. Return alone cannot answer that, so the evaluation stage runs every signal through the same backtester, over the same window, with the same costs, and compares it with simply holding SPY.

## Objective

Build `python -m pipeline evaluate` to compute total return, CAGR, volatility, Sharpe, Sortino, maximum drawdown, Calmar, win rate, profit factor, exposure, turnover, alpha and beta for every signal against a buy-and-hold benchmark, and interpret the ranking.

## Before You Start

- Part A: start the Trading Simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000 > Course Labs > Lab 09. Install steps: Learner Guide, 'Download, Install and Start the Trading Simulator Software'.
- Part B: open a terminal in the labs folder, activate your virtual environment and open your AI coding assistant there; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

## Part A - In the Trading Simulator

*Start the simulator (uv run trader serve --with-worker, or docker compose up -d) and open http://localhost:8000. In the sidebar open Course Labs > Lab 09: the app shows the same steps with a Go*

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*button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.*

### A1. Open the analytics

Analytics: choose the EMA 10/30 seeded backtest. Hover every metric for its formula.

Open: http://localhost:8000/#/analytics  (Analytics)

*Check: You can define Sharpe, Sortino, Calmar, max drawdown, profit factor, expectancy, exposure and turnover.*

### A2. Compare with buy-and-hold

Tick the EMA backtest and 'Buy and hold SPY' and press Compare.

Open: http://localhost:8000/#/analytics

*Check: Buy-and-hold has higher CAGR but a deeper drawdown; the strategy has lower exposure.*

### A3. Glossary

Open Help & Glossary > Metric glossary.

Open: http://localhost:8000/#/help?tab=glossary

*Check: Each formula matches the course slides.*

**Expected in the app: A metrics table you can explain line by line.**

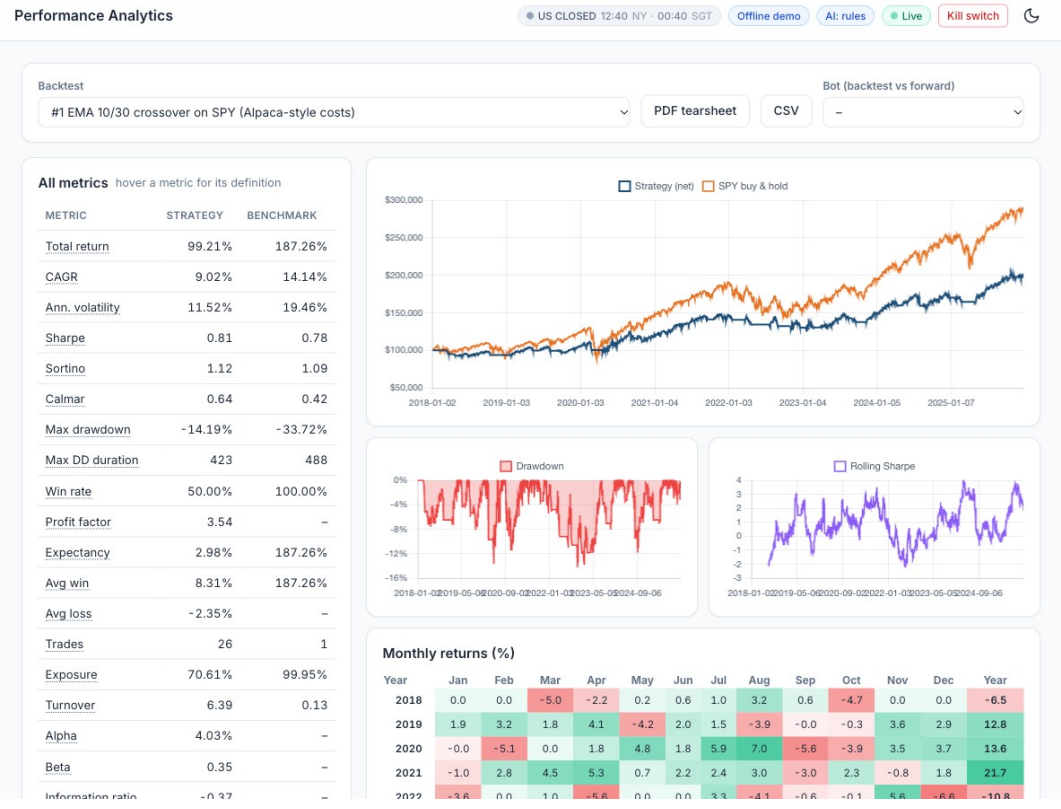



*Figure: Lab 09 in the Trading Simulator (Lab09_analytics.png).*

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## Part B - In Code with Your AI Assistant

*The same pipeline stage in code. AI writes the code; you own the algorithm: paste each prompt into your AI coding assistant in the labs folder (the finished reference code is already in labs/Lab09-performance-metrics/), run it, and check the result against Part A. The Practical Performance assessment uses this code path.*

### Step 1: Read the metric formulas (K3)

Open common/metrics.py; each function's docstring gives its formula (annualised with 252 trading days).

| AI prompt - paste into your AI coding assistant |
|---|
| Explain each function in labs/common/metrics.py with its formula and one sentence on what a portfolio manager learns from it; say which metrics need trade-level returns and which need a benchmark. Do not change the file. |

*Check: You can explain the difference between Sharpe (all volatility) and Sortino (downside only).*

### Step 2: Build the evaluation stage (A3, K3)

Ask your assistant for pipeline/evaluate.py.

| AI prompt - paste into your AI coding assistant |
|---|
| Create labs/pipeline/evaluate.py. evaluate_strategy(bars, name, costs, start, bench_ret, **params) backtests pipeline.signals.target_position(name, bars, **params) with common.backtest.backtest (next-bar open fills, costs = pipeline.config.cost_model(cfg)), slices returns and position from `start` and returns common.metrics.summary(...) with trade net returns and the benchmark. run(symbol='SPY'): start = first date the 200-day SMA exists (so every signal is warm), benchmark = buy_and_hold sliced from start; evaluate every signal in SIGNALS; print a table (rows = strategies + buy & hold, columns CAGR, Ann. volatility, Sharpe, Sortino, Max drawdown, Calmar, Trades, Win rate, Profit factor, Exposure, Turnover, Alpha, Beta) ranked by Sharpe and formatted with common.metrics.format_table; list which strategies beat buy and hold on Sharpe and how many have a smaller drawdown; print 'Sharpe by hand' for trend_filter (mean / stdev x sqrt(252)), the worst day and SPY's drawdown trough; save _outputs/Lab09_metrics.csv and chart Lab09_metrics.png (top 4 by Sharpe plus buy & hold, equity and drawdown). Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow. |

*Check: python -m pipeline evaluate --help shows --symbol.*

### Step 3: Run the evaluation (A3)

Run: python -m pipeline evaluate --symbol SPY (same as: python3 Lab09-performance-metrics/evaluate_stage.py).

*Check: Window 2018-10-16 to 2025-12-31, Alpaca-style costs. ema_cross: CAGR 11.09%, vol 11.87%, Sharpe 0.95, max drawdown -14.19%, Calmar 0.78, 26 trades, win rate 50.00%, profit factor 3.54, beta 0.35. SPY buy & hold: 15.28%, 19.93%, 0.81, -33.72%, 0.45. trend_filter: 9.70%, 12.42%, 0.81, -19.36%, 24 trades, win 41.67%, profit factor 3.79, exposure 78.53%, beta 0.38. Only ema_cross beats buy & hold on Sharpe; 7 of 8 have a smaller drawdown.*

### Step 4: Check one metric by hand (K3)

Read the 'Sharpe by hand' line and confirm it equals the table.

*Check: mean 0.00040 / stdev 0.00782 x sqrt(252) = 0.81, the same as trend_filter's Sharpe in the table.*

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### Step 5: Interpret trade and market metrics (A3, K3)

Explain why trend_filter can win only 41.67% of trades yet have a profit factor of 3.79, and why beta is 0.38 when exposure is 78.53%.

| AI prompt - paste into your AI coding assistant |
| --- |
| Using labs/_outputs/Lab09_metrics.csv, write a 120-word note for Lion City Capital's investment committee comparing ema_cross, trend_filter and SPY buy & hold on return, risk and risk-adjusted return. Quote the numbers, explain win rate versus profit factor, and do not recommend live trading. |

*Check: Few large winners pay for many small losers; the strategy is out of the market in many stress periods, so its beta is far below its exposure.*

### Step 6: Locate the worst periods (K3)

Read the worst-day and drawdown-trough line and find those dates on _outputs/Lab09_metrics.png.

*Check: Worst trend_filter day 2020-06-11 (-5.76%); worst SPY day 2020-03-16 (-10.94%); SPY drawdown trough 2020-03-23.*

### Step 7: Add a metric to the stage (K3)

Extend the evaluation table with one more risk metric.

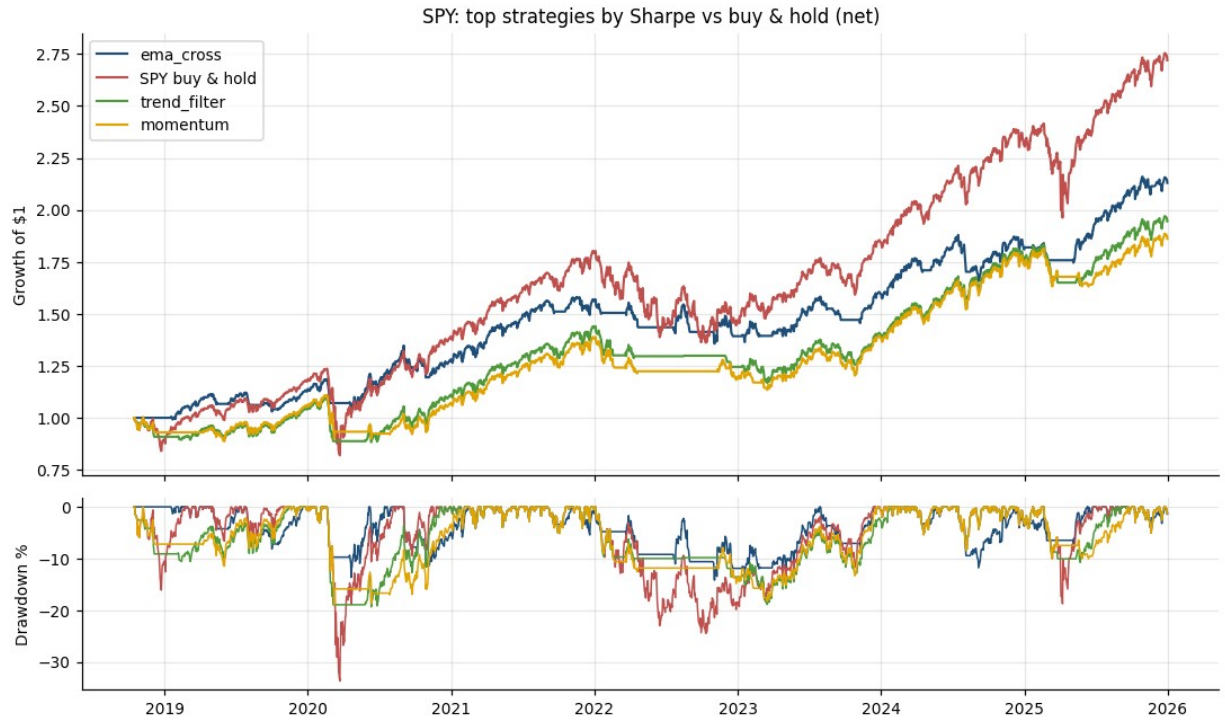
| AI prompt - paste into your AI coding assistant |
| --- |
| In labs/pipeline/evaluate.py add an 'Ulcer index' column = sqrt(mean(drawdown% squared)) computed from common.metrics.drawdown_series of each strategy's returns, without changing common/metrics.py. Show it in the printed table and the CSV, and explain in one sentence what it adds to max drawdown. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow. |

*Check: The new column shows a much lower Ulcer index for ema_cross than for SPY buy & hold.*

## Expected Output

python -m pipeline evaluate --symbol SPY: window 2018-10-16 to 2025-12-31; ema_cross CAGR 11.09%, Sharpe 0.95, max drawdown -14.19% (best Sharpe); SPY buy & hold 15.28%, 0.81, -33.72%; trend_filter 9.70%, 0.81, -19.36%, 24 trades, win rate 41.67%, profit factor 3.79, beta 0.38; bollinger_breakout last (Sharpe 0.35); files _outputs/Lab09_metrics.csv and chart _outputs/Lab09_metrics.png.

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*Lab 9 output (Lab09_metrics.png), offline mode on the bundled data.*

## Debrief

Performance metrics (K3) answer different questions: CAGR measures growth, Sharpe and Sortino reward per unit of risk, drawdown and Calmar the pain an investor must survive, beta how much is just the market. Running every strategy through one evaluation stage (A3) is what makes the comparison fair.

## Troubleshooting

| Symptom | What to do |
|---|---|
| Sharpe looks too high | Check that daily (not cumulative) returns are used and annualised with sqrt(252). |
| Win rate below 50% yet profitable | Expected for trend following: few large winners pay for many small losers (see profit factor). |
| Strategies cover different periods | Slice every return series from the same start date (first valid 200-day SMA). |

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